

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Exercise

Binomial Theorem

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If the coefficients of x^7 & x^8 in the expansion of $\left[2 + \frac{x}{3}\right]^n$ are equal, then the value of n is :

(A) 15 (B) 45 (C) 55 (D) 56

BT0001

2. Set of value of r for which, ${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ contains :

(A) 4 element (B) 5 elements (C) 7 elements (D) 10 elements

BT0002

3. If the constant term of the binomial expansion $\left(2x - \frac{1}{x}\right)^n$ is -160 , then n is equal to -

(A) 4 (B) 6 (C) 8 (D) 10

BT0003

4. The coefficient of x^{49} in the expansion of $(x-1)\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2^2}\right) \dots \left(x - \frac{1}{2^{49}}\right)$ is equal to -

(A) $-2\left(1 - \frac{1}{2^{50}}\right)$

(B) +ve coefficient of x

(C) -ve coefficient of x

(D) $-2\left(1 - \frac{1}{2^{49}}\right)$

BT0004

5. The largest real value for x such that $\sum_{k=0}^4 \left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right) = \frac{8}{3}$ is -

(A) $2\sqrt{2} - 5$

(B) $2\sqrt{2} + 5$

(C) $-2\sqrt{2} - 5$

(D) $-2\sqrt{2} + 5$

BT0005

6. The expression $[x + (x^3 - 1)^{1/2}]^5 + [x - (x^3 - 1)^{1/2}]^5$ is a polynomial of degree

(A) 5

(B) 6

(C) 7

(D) 8

BT0006

7. Number of rational terms in the expansion of $(\sqrt{2} + \sqrt[4]{3})^{100}$ is :

(A) 25

(B) 26

(C) 27

(D) 28

BT0007

8. Given $(1 - 2x + 5x^2 - 10x^3)(1 + x)^n = 1 + a_1x + a_2x^2 + \dots$ and that $a_1^2 = 2a_2$ then the value of n is-

(A) 6

(B) 2

(C) 5

(D) 3

BT0008

9. The sum of the co-efficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is -
 (A) 2.6^{10} (B) 3.6^{10} (C) 6^{11} (D) none

BT0009

10. Co-efficient of α^t in the expansion of,
 $(\alpha + p)^{m-1} + (\alpha + p)^{m-2}(\alpha + q) + (\alpha + p)^{m-3}(\alpha + q)^2 + \dots(\alpha + q)^{m-1}$ where $\alpha \neq -q$ and $p \neq q$ is :
 (A) $\frac{{}^m C_t (p^t - q^t)}{p - q}$ (B) $\frac{{}^m C_t (p^{m-t} - q^{m-t})}{p - q}$ (C) $\frac{{}^m C_t (p^t + q^t)}{p - q}$ (D) $\frac{{}^m C_t (p^{m-t} + q^{m-t})}{p - q}$

BT0010

11. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the sum

$$\binom{99}{97} + \binom{98}{96} + \binom{97}{95} + \dots + \binom{3}{1} + \binom{2}{0} \text{ equals -}$$

- (A) $\binom{99}{97}$ (B) $\binom{100}{98}$ (C) $\binom{99}{98}$ (D) $\binom{100}{97}$

BT0011

12. If $n \in \mathbb{N}$ & n is even, then $\frac{1}{1.(n-1)!} + \frac{1}{3!.(n-3)!} + \frac{1}{5!.(n-5)!} + \dots + \frac{1}{(n-1)!1!} =$

- (A) 2^n (B) $\frac{2^{n-1}}{n!}$ (C) $2^n n!$ (D) none of these

BT0013

13. If $|x| < 1$, then the co-efficient of x^n in the expansion of $(1 + x + x^2 + x^3 + \dots)^2$ is
 (A) n (B) $n - 1$ (C) $n + 2$ (D) $n + 1$

BT0109

14. The co-efficient of x^4 in the expansion of $(1 - x + 2x^2)^{12}$ is
 (A) ${}^{12}C_3$ (B) ${}^{13}C_3$ (C) ${}^{14}C_4$ (D) ${}^{12}C_3 + 3 {}^{13}C_3 + {}^{14}C_4$

BT0110

15. The value of the expression ${}^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3$ is equal to

- (A) ${}^{47}C_5$ (B) ${}^{52}C_5$ (C) ${}^{52}C_4$ (D) ${}^{49}C_4$

BT0111

16. The value of $\binom{50}{0}\binom{50}{1} + \binom{50}{1}\binom{50}{2} + \dots + \binom{50}{49}\binom{50}{50}$ is, where ${}^nC_r = \binom{n}{r}$

- (A) $\binom{100}{50}$ (B) $\binom{100}{51}$ (C) $\binom{50}{25}$ (D) $\binom{50}{25}^2$

BT0112

17. The last three digits in $10!$ are :

- (A) 800 (B) 700 (C) 500 (D) 600

BT0113

18. $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} =$

- (A) $\frac{n}{2}$ (B) $\frac{n+1}{2}$ (C) $(n+1)\frac{n}{2}$ (D) $\frac{n(n-1)}{2(n+1)}$

BT0115

19. Find numerically greatest term in the expansion of $(2+3x)^9$, when $x = 3/2$

- (A) ${}^9C_6 \cdot 2^9 \cdot (3/2)^{12}$ (B) ${}^9C_3 \cdot 2^9 \cdot (3/2)^6$ (C) ${}^9C_5 \cdot 2^9 \cdot (3/2)^{10}$ (D) ${}^9C_4 \cdot 2^9 \cdot (3/2)^8$

BT0116

20. The greatest integer less than or equal to $(\sqrt{2}+1)^6$ is

- (A) 196 (B) 197 (C) 198 (D) 199

BT0117

21. If $k \in \mathbb{R}^+$ and the middle term of $\left(\frac{k}{2} + 2\right)^8$ is 1120, then value of k is

- (A) 3 (B) 2 (C) 1 (D) 4

BT0118

22. If the second term of the expansion $\left[a^{1/13} + \frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$ is

- (A) 4 (B) 3 (C) 12 (D) 6

BT0119

23. In the expansion of $(7^{1/3} + 11^{1/9})^{6561}$, the number of terms free from radicals is

- (A) 730 (B) 729 (C) 725 (D) 750

BT0120

24. The value of m , for which the coefficients of the $(2m+1)^{\text{th}}$ and $(4m+5)^{\text{th}}$ terms in the expansion of $(1+x)^{10}$ are equal to, is

- (A) 3 (B) 1 (C) 5 (D) 8

BT0121

25. Greatest term in the binomial expansion of $(a + 2x)^9$ when $a = 1$ & $x = \frac{1}{3}$ is :

- (A) 3rd & 4th (B) 4th & 5th (C) only 4th (D) only 5th

BT0019

26. Let $(5 + 2\sqrt{6})^n = p + f$ where $n, p \in \mathbb{N}$ and $0 < f < 1$ then the value of $f^2 - f + pf - p$ is -

- (A) a natural number (B) a negative integer (C) a prime number (D) are irrational number

BT0020

27. If $\sum_{r=1}^{10} r(r-1) {}^{10}C_r = k \cdot 2^9$, then k is equal to -

- (A) 10 (B) 45 (C) 90 (D) 100

BT0022

28. The sum $\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \dots + \frac{\binom{11}{11}}{12}$ equals $\left(\text{where } \binom{n}{r} \text{ denotes } {}^nC_r \right)$

- (A) $\frac{2^{11}}{12}$ (B) $\frac{2^{12}}{12}$ (C) $\frac{2^{11}-1}{12}$ (D) $\frac{2^{12}-1}{12}$

BT0023

29. **Statement-1 :** The sum of the series ${}^nC_0 \cdot {}^mC_r + {}^nC_1 \cdot {}^mC_{r-1} + {}^nC_2 \cdot {}^mC_{r-2} + \dots + {}^nC_r \cdot {}^mC_0$ is equal to ${}^{n+m}C_r$, where nC_r 's and mC_r 's denotes the combinatorial coefficients in the expansion of $(1+x)^n$ and $(1+x)^m$ respectively.

Statement-2 : Number of ways in which r children can be selected out of $(n+m)$ children consisting of n boys and m girls if each selection may consist of any number of boys and girls is equal to ${}^{n+m}C_r$.

- (A) Statement-1 is true, statement-2 is true ; statement-2 is a correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true ; statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

BT0024

30. Let ${}^0C_0 = 1$, then $\sum_{m=0}^n \sum_{p=0}^m {}^nC_m \cdot {}^mC_p$ is equal to

- (A) $2^n - 1$ (B) 3^n (C) $3^n - 1$ (D) 2^n

BT0123

EXERCISE (O-2)

Multiple Correct Answer Type

1. If it is known that the third term of the binomial expansion $(x + x^{\log_{10} x})^5$ is 10^6 then x is equal to-

(A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5

BT0014

2. In the expansion of $(x^3 + 3.2^{-\log_{\sqrt{2}} \sqrt{x^3}})^{11}$

(A) there appears a term with the power x^2
 (B) there does not appear a term with the power x^2
 (C) there appears a term with the power x^{-3}
 (D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$

BT0015

3. In the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$, the term which does not contain x is -

(A) ${}^{11}C_4 - {}^{10}C_3$ (B) ${}^{10}C_7$ (C) ${}^{10}C_4$ (D) ${}^{11}C_5 - {}^{10}C_5$

BT0016

4. Let $(1+x^2)^2(1+x)^n = A_0 + A_1x + A_2x^2 + \dots$. If A_0, A_1, A_2 are in A.P. then the value of n is -

(A) 2 (B) 3 (C) 5 (D) 7

BT0017

5. Consider $E = \left(\sqrt[8]{x} + \sqrt[5]{y}\right)^z = I + f, 0 \leq f < 1, I \in \mathbb{Z}$

(A) If $x = 5, y = 2, z = 100$, then number of irrational terms in expansion of E is 98
 (B) If $x = 5, y = 2, z = 100$, then number of rational terms in expansion of E is 4
 (C) If $x = 16, y = 1$ & $z = 6$, then $I = 197$
 (D) If $x = 16, y = 1$ & $z = 6$, then $f = (\sqrt{2} - 1)^6$

BT0018

6. If $(9 + \sqrt{80})^n = I + f$ where I, n are integers and $0 < f < 1$, then -

(A) I is an odd integer (B) I is an even integer
 (C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$

BT0021

7. Which of the following statement(s) is/are correct ?

(A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$

(B) Integral part of $(9 + 4\sqrt{5})^n$, $n \in \mathbb{N}$ is even.

(C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$

(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x| < \frac{2}{3}$.

BT0025

8. If for $n \in \mathbb{I}$, $n > 10$; $1 + (1+x) + (1+x)^2 + \dots + (1+x)^n = \sum_{k=0}^n a_k \cdot x^k$, $x \neq 0$ then

(A) $\sum_{k=0}^n a_k = 2^{n+1}$

(B) $a_{n-2} = \frac{n(n+1)}{2}$

(C) $a_p > a_{p-1}$ for $p < \frac{n}{2}$, $p \in \mathbb{N}$

(D) $(a_9)^2 - (a_8)^2 = {}^{n+2}C_{10} ({}^{n+1}C_{10} - {}^{n+1}C_9)$

BT0026

9. Let $P(n) = \sum_{r=0}^n \frac{(-1)^r}{r+1} {}^nC_r$. Now which of the following holds good ?

(A) $|P_{10}|$ is harmonic mean of $|P_9|$ & $|P_{11}|$

(B) $\sum_{r=5}^{10} P(r)P(r-1) = -\frac{6}{55}$

(C) $|P_{10}|$ is arithmetic mean of $|P_9|$ & $|P_{11}|$

(D) $\sum_{r=5}^{10} P(r)P(r-1) = \frac{6}{55}$

BT0027

10. Let $(1+x)^m = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_mx^m$, where $C_r = {}^mC_r$ and $A = C_1C_3 + C_2C_4 + C_3C_5 + C_4C_6 + \dots + C_{m-2}C_m$, then -

(A) $A \geq {}^{2m}C_{m-2}$

(B) $A < {}^{2m}C_{m-2}$

(C) $A > C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$

(D) $A < C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$

BT0028

11. In the expansion $\left(\sqrt[3]{4} + \frac{1}{\sqrt[4]{6}}\right)^{20}$

(A) The number of irrational terms is 19

(B) Middle term is irrational

(C) The number of rational terms is 2

(D) 9th term is rational

BT0125

12. The coefficient of x^4 in $\left(\frac{1+x}{1-x}\right)^2, |x| < 1$, is

- (A) 4 (B) -4 (C) $10 + {}^4C_2$ (D) 16

BT0126

13. $7^9 + 9^7$ is divisible by :

- (A) 16 (B) 24 (C) 64 (D) 72

BT0127

14. The sum of the series $\sum_{r=1}^n (-1)^{r-1} \cdot {}^nC_r (a-r)$ is equal to

- (A) 5 if $a = 5$ (B) -5 if $a = 5$ (C) -5 if $a = -5$ (D) 5 if $a = -5$

BT0128

15. Let $a_n = \frac{1000^n}{n!}$ for $n \in \mathbb{N}$, then a_n is greatest, when

- (A) $n = 997$ (B) $n = 998$ (C) $n = 999$ (D) $n = 1000$

BT0129

16. ${}^nC_0 - 2 \cdot {}^nC_1 + 3 \cdot {}^nC_2 - 4 \cdot {}^nC_3 + \dots + (-1)^n (n+1) {}^nC_n 3^n$ is equal to

- (A) $2^n \left(\frac{3n}{2} + 1 \right)$ if n is even (B) $2^n \left(n + \frac{3}{2} \right)$ if n is even
(C) $-2^n \left(\frac{3n}{2} + 1 \right)$ if n is odd (D) $2^n \left(n + \frac{3}{2} \right)$ if n is odd

BT0130

17. The expansion of $(3x+2)^{-1/2}$ is valid in ascending powers of x , if x lies in the interval

- (A) $(0, 2/3)$ (B) $(-3/3, 3/2)$ (C) $(-2/3, 2/3)$ (D) $(-\infty, -3/2) \cup (3/2, \infty)$

BT0132

18. If $(1+2x+3x^2)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{20}x^{20}$, then

- (A) $a_1 = 20$ (B) $a_2 = 210$ (C) $a_4 = 8085$ (D) $a_{20} = 2^2 \cdot 3^7 \cdot 7$

BT0133

19. In the expansion of $(x+y+z)^{25}$

- (A) every term is of the form ${}^{25}C_r \cdot {}^rC_k \cdot x^{25-r} \cdot y^{r-k} \cdot z^k$ (B) the coefficient of $x^8y^9z^9$ is 0
(C) the number of terms is 325 (D) none of these

BT0134

20. If $(1+x+2x^2)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$, then $a_0 + a_2 + a_4 + \dots + a_{38}$ is equal to

- (A) $2^{19}(2^{30} + 1)$ (B) $2^{19}(2^{20} - 1)$ (C) $2^{39} - 2^{19}$ (D) $2^{39} + 2^{19}$

BT0135

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Consider, sum of the series $\sum_{0 \leq i < j \leq n} f(i, j)$

In the given summation, i and j are not independent.

In the sum of series $\sum_{i=0}^n \sum_{j=0}^n f(i, j)$, i and j are independent. In this summation, three types of terms occur, those when $i < j$, $i > j$ and $i = j$.

$$\sum_{i=0}^n \sum_{j=0}^n f(i, j) = \sum_{0 \leq i < j \leq n} f(i, j) + \sum_{0 \leq j < i \leq n} f(i, j) + \sum_{k=0}^n f(k, k)$$

Also if $f(i, j) = f(j, i)$ then sum of terms when $i < j$ is equal to the sum of the terms when $i > j$. so, in this case

$$\sum_{i=0}^n \sum_{j=0}^n f(i, j) = 2 \sum_{0 \leq i < j \leq n} f(i, j) + \sum_{k=0}^n f(k, k)$$

$$\Rightarrow \sum_{0 \leq i < j \leq n} f(i, j) = \frac{\sum_{i=0}^n \sum_{j=0}^n f(i, j) - \sum_{k=0}^n f(k, k)}{2}$$

When $f(i)$ and $f(j)$ are not symmetrical, we find the sum by listing all the terms.

1. $\sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j$ is equal to

(A) $\frac{2^{2n} - {}^nC_n}{2}$ (B) $\frac{2^{2n} + {}^nC_n}{2}$ (C) $\frac{2^{2n} - {}^nC_n}{2}$ (D) $\frac{2^{2n} + {}^nC_n}{2}$

BT0122

2. The value of $\sum_{0 \leq r < s \leq n} (r+s)({}^nC_r + {}^nC_s + {}^nC_r {}^nC_s)$ is equal to

(A) $n^2 \cdot 2^n - \frac{n}{2} [2^{2n} - {}^{2n}C_n]$ (B) $n^2 \cdot 2^n + \frac{n}{2} [2^{2n} - {}^{2n}C_n]$
 (C) $n^2 \cdot 2^n - \frac{n}{2} [2^{2n} + {}^{2n}C_n]$ (D) $n^2 \cdot 2^n + \frac{n}{2} [2^n - 1]$

BT0136

Paragraph for Question 3 to 5

If $n \in \mathbb{N}$ and if $(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r$, where $a_0, a_1, a_2, \dots, a_{2n}$ are real numbers.

3. The unit place of $2 \sum_{r=0}^n a_{2r}$ can be

(A) 0 (B) 2 (C) 4 (D) 8

BT0012

4. The unit place of $2 \sum_{r=1}^n a_{2r-1}$ can be

(A) 0 (B) 2 (C) 4 (D) 8

BT0012

5. The value of a_{2n-1} is divisible by

(A) 2^{2n} (B) $n \cdot 2^{2n}$ (C) $(n-1)2^{2n}$ (D) $(n+1)2^{2n}$

BT0012

Paragraph for Question 6 to 8

Let us consider the binomial expression $(1+x)^n = \sum_{r=0}^n a_r x^r$, where a_1, a_2, a_3 are in A.P.

Consider another binomial expression $A = (2^{1/3} + 3^{1/4})^{14n}$. The expansion of A contains some rational terms $T_{b_1}, T_{b_2}, \dots, T_{b_m}$ ($b_1 < b_2 < b_3 < \dots < b_m$), where n is greater than the even prime number.

6. The value of $a_1 + a_2 + a_3$ is -

BT0137

7. The value of $\sum_{r=0}^n a_r$ is -

BT0138

8. The value of b_m is -

BT0139

Matrix Match Type

- | | | |
|-----------|---------------|----------------|
| 9. | List-I | List-II |
|-----------|---------------|----------------|

(A) If $\sum_{r=0}^{10} (r+2)^r {}^{10}C_r = 10 \cdot 2^a + 3^b$, then $(a+b)$ is equal to	(P) 7
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(B) If 4 th term in the expansion of $(x + (ax)^{-1})^m$, $m \in \mathbb{N}$ is $\frac{5}{2}$, then ma is equal to	(Q) 12
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(C) Let λ be the value of numerically greatest term in the expansion of $(1+5x)^{10}$ at $x=1$, then $\frac{\lambda}{{}^{10}C_9 \cdot 5^7}$ is equal to	(R) 19
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(D) If ${}^nP_r = 5040({}^{n-1}C_r + {}^{n-1}C_{r-1})$, then r is equal to	(S) 25
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(A) (A) $\rightarrow$ (R), (B) $\rightarrow$ (Q), (C) $\rightarrow$ (S), (D) $\rightarrow$ (P)

(B) (A) $\rightarrow$ (R), (B) $\rightarrow$ (Q), (C) $\rightarrow$ (P), (D) $\rightarrow$ (S)

(C) (A) $\rightarrow$ (Q), (B) $\rightarrow$ (R), (C) $\rightarrow$ (S), (D) $\rightarrow$ (P)

(D) (A) $\rightarrow$ (R), (B) $\rightarrow$ (S), (C) $\rightarrow$ (Q), (D) $\rightarrow$ (P)

BT0140

10. Column-I

Column-II

- (A) The coefficient of x^{-9} in the expansion of $\left(\frac{x^2}{2} + \frac{2}{x}\right)^9$ is (P) 1532
- (B) The term independent of x in the expansion of $\left(2x + \frac{3}{x}\right)^6$ is (Q) 476
- (C) The coefficient of x^{10} in the expansion of $(1 + x^2 - x^3)^8$ is (R) 512
- (D) The coefficient of x^4 in the expansion of $(3x + 1)^7(1 - 2x + 3x^2)$ is (S) 4320

BT0141

EXERCISE (O-4)

Numerical Grid Type

- Let a and b be the coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$ and $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ respectively, then value of $(a - b)$ is
BT0063
- Let $a = (4^{1/401} - 1)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + \dots + {}^nC_n \cdot a^{n-1}$, then the value of $\frac{(b_{2006} - b_{2005})}{512}$ is
BT0066
- Let $P = (2 + \sqrt{3})^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function, then the value of $\left(\frac{f^2}{1-f}\right) - 720$ is
BT0068
- If $(7 + 4\sqrt{3})^n = p + \beta$ where n & p are positive integers and β is a proper fraction then value of $(1 - \beta)(p + \beta) - 1$ is
BT0069
- The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$
BT0102
- The index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has uniquely numerically greatest coefficient ($n \in \mathbb{N}$), is
BT0064
- The sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$, is
BT0065
- The coefficient of x^{49} in the polynomial $\left(x - \frac{C_1}{C_0}\right)\left(x - 2^2 \cdot \frac{C_2}{C_1}\right)\left(x - 3^2 \cdot \frac{C_3}{C_2}\right) \dots \left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right)$, where $C_r = {}^{50}C_r$, is
BT0070
- The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is :-
BT0076
- Let $f(x) = 1 - x + x^2 - x^3 + \dots + x^{16} - x^{17} = a_0 + a_1(1 + x) + a_2(1 + x)^2 + \dots + a_{17}(1 + x)^{17}$, then the value of a_2 is
BT0036

EXERCISE (JM)

1. The sum of coefficients of integral powers of x in the binomial expansion of $(1 - 2\sqrt{x})^{50}$ is : [JEE(Main)-2015]
 (1) $\frac{1}{2}(3^{50} - 1)$ (2) $\frac{1}{2}(2^{50} + 1)$ (3) $\frac{1}{2}(3^{50} + 1)$ (4) $\frac{1}{2}(3^{50})$
BT0080
2. If the number of terms in the expansion of $\left(1 - \frac{2}{x} + \frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is :- [JEE(Main)-2016]
 (1) 729 (2) 64 (3) 2187 (4) 243
BT0081
3. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ is :- [JEE(Main)-2017]
 (1) $2^{20} - 2^{10}$ (2) $2^{21} - 2^{11}$ (3) $2^{21} - 2^{10}$ (4) $2^{20} - 2^9$
BT0082
4. The sum of the co-efficients of all odd degree terms in the expansion of $(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$, ($x > 1$) is - [JEE(Main)-2018]
 (1) 0 (2) 1 (3) 2 (4) -1
BT0083
5. If the fractional part of the number $\frac{2^{403}}{15}$ is $\frac{k}{15}$, then k is equal to : [JEE(Main)- 2019]
 (1) 14 (2) 6 (3) 4 (4) 8
BT0084
6. The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is [JEE(Main)- 2019]
 (1) 12 (2) 15 (3) 10 (4) 14
BT0085
7. If $\sum_{r=0}^{25} \left\{ {}^{50}C_r \cdot {}^{50-r}C_{25-r} \right\} = K \left({}^{50}C_{25} \right)$, then K is equal to : [JEE(Main)- 2019]
 (1) $2^{25} - 1$ (2) $(25)^2$ (3) 2^{25} (4) 2^{24}
BT0086
8. The sum of the real values of x for which the middle term in the binomial expansion of $\left(\frac{x^3}{3} + \frac{3}{x}\right)^8$ equals 5670 is : [JEE(Main)- 2019]
 (1) 6 (2) 8 (3) 0 (4) 4
BT0087

9. The value of r for which ${}^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \dots + {}^{20}C_0 \cdot {}^{20}C_r$ is maximum, is [JEE(Main)- 2019]
- (1) 20 (2) 15 (3) 11 (4) 10

BT0088

10. Let $(x + 10)^{50} + (x - 10)^{50} = a_0 + a_1x + a_2x^2 + \dots + a_{50}x^{50}$, for all $x \in \mathbb{R}$, then $\frac{a_2}{a_0}$ is equal to :- [JEE(Main)- 2019]
- (1) 12.50 (2) 12.00 (3) 12.75 (4) 12.25

BT0089

11. Let $S_n = 1 + q + q^2 + \dots + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$, where q is a real number and $q \neq 1$. If ${}^{101}C_1 + {}^{101}C_2 \cdot S_1 + \dots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$, then α is equal to :- [JEE(Main)- 2019]
- (1) 2^{100} (2) 200 (3) 2^{99} (4) 202

BT0090

12. The total number of irrational terms in the binomial expansion of $(7^{1/5} - 3^{1/10})^{60}$ is : [JEE(Main)- 2019]
- (1) 55 (2) 49 (3) 48 (4) 54

BT0091

13. If some three consecutive in the binomial expansion of $(x + 1)^n$ is powers of x are in the ratio $2 : 15 : 70$, then the average of these three coefficient is :- [JEE(Main)- 2019]
- (1) 964 (2) 625 (3) 227 (4) 232

BT0092

14. The coefficient of x^{18} in the product $(1 + x)(1 - x)^{10}(1 + x + x^2)^9$ is : [JEE(Main)- 2019]
- (1) -84 (2) 84 (3) 126 (4) -126

BT0093

15. If ${}^{20}C_1 + (2^2) \cdot {}^{20}C_2 + (3^2) \cdot {}^{20}C_3 + \dots + (20^2) \cdot {}^{20}C_{20} = A(2^\beta)$, then the ordered pair (A, β) is equal to : [JEE(Main)- 2019]
- (1) (420, 18) (2) (380, 19) (3) (380, 18) (4) (420, 19)

BT0094

16. The term independent of x in the expansion of $\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$ is equal to : [JEE(Main)- 2019]
- (1) 36 (2) -108 (3) -72 (4) -36

BT0095

17. The coefficient of x^7 in the expression $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + \dots + x^{10}$ is :
 [JEE(Main)- 2020]
 (1) 120 (2) 330 (3) 210 (4) 420
BT0096
18. If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+\dots+x^{2n})(1-x+x^2-x^3+\dots+x^{2n})$ is 61, then n is equal to _____.
 [JEE(Main)- 2020]
BT0097
19. If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x+\sqrt{x^2-1}\right)^6 + \left(x-\sqrt{x^2-1}\right)^6$, then
 [JEE(Main)- 2020]
 (1) $\alpha + \beta = 60$ (2) $\alpha + \beta = -30$ (3) $\alpha - \beta = -132$ (4) $\alpha - \beta = 60$
BT0098
20. In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$, if ℓ_1 is the least value of the term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ and ℓ_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, then the ratio $\ell_2 : \ell_1$ is equal to :
 [JEE(Main)- 2020]
 (1) 1 : 8 (2) 1 : 16 (3) 8 : 1 (4) 16 : 1
BT0099
21. If $C_r \equiv {}^{25}C_r$ and $C_0 + 5.C_1 + 9.C_2 + \dots + (101).C_{25} = 2^{25}.k$, then k is equal to _____.
 [JEE(Main)- 2020]
BT0100
22. If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^nC_2 + {}^{n+1}C_2 + {}^{n+2}C_2 + \dots + {}^{2n}C_2)$ is :
 [JEE(Main)- 2021]
 (1) $\frac{n(n-1)(2n+1)}{6}$ (2) $\frac{n(n+1)(2n+1)}{6}$ (3) $\frac{n(2n+1)(3n+1)}{6}$ (4) $\frac{n(n+1)^2(n+2)}{12}$
BT0142
23. If the remainder when x is divided by 4 is 3, then the remainder when $(2020+x)^{2022}$ is divided by 8 is _____.
 [JEE(Main)- 2021]
BT0143
24. The remainder when $(2021)^{2023}$ is divided by 7 is :
 [JEE(Main)- 2022]
 (A) 1 (B) 2 (C) 5 (D) 6
BT0144
25. If the sum of the coefficients of all the positive powers of x , in the binomial expansion of $\left(x^n + \frac{2}{x^5}\right)^7$ is 939, then the sum of all the possible integral values of n is :
 [JEE(Main)- 2022]
BT0145
26. If $({}^{40}C_0) + ({}^{41}C_1) + ({}^{42}C_2) + \dots + ({}^{60}C_{20}) = \frac{m}{n} {}^{60}C_{20}$, m and n are coprime, then $m+n$ is equal to _____.
 [JEE(Main)- 2022]
BT0146

EXERCISE (JA)

- Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is -
 (A) 1051 (B) 1106 (C) 1113 (D) 1120
[JEE(Advanced)-2014, 3(-1)]
BT0103
- The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is **[JEE 2015, 4M, -0M]**
BT0104
- Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)^{51}C_3$ for some positive integer n . Then the value of n is **[JEE(Advanced)-2016, 3(0)]**
BT0105
- Let $X = \binom{10}{1}C_1^2 + 2\binom{10}{2}C_2^2 + 3\binom{10}{3}C_3^2 + \dots + 10\binom{10}{10}C_{10}^2$, where $\binom{10}{r}C_r$, $r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is _____. **[JEE(Advanced)-2018, 3(0)]**
BT0106
- Suppose $\det \begin{bmatrix} \sum_{k=0}^n k & \sum_{k=0}^n {}^nC_k k^2 \\ \sum_{k=0}^n {}^nC_k k & \sum_{k=0}^n {}^nC_k 3^k \end{bmatrix} = 0$, holds for some positive integer n . Then $\sum_{k=0}^n \frac{{}^nC_k}{k+1}$ equals **[JEE(Advanced)-2019, 3(0)]**
BT0107
- For non-negative integers s and r , let $\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \leq s, \\ 0 & \text{if } r > s. \end{cases}$ For positive integers m and n , let $g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{\binom{n+p}{p}}$ where for any nonnegative integer p , $f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \binom{n+i}{p} \binom{p+n}{p-i}$
 Then which of the following statements is/are TRUE ? **[JEE(Advanced)-2020]**
 (A) $g(m, n) = g(n, m)$ for all positive integers m, n
 (B) $g(m, n+1) = g(m+1, n)$ for all positive integers m, n
 (C) $g(2m, 2n) = 2g(m, n)$ for all positive integers m, n
 (D) $g(2m, 2n) = (g(m, n))^2$ for all positive integers m, n
BT0108
- Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} is equal to the coefficient of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of $2b$ is **[JEE(Advanced)-2023]**
BT0147

ANSWER KEY

Do yourself - 1

1. ${}^5C_0 x(3x^2)^5 + {}^5C_1 (3x^2)^4 \left(-\frac{x}{2}\right) + {}^5C_2 (3x^2)^3 \left(-\frac{x}{2}\right)^2 + {}^5C_3 (3x^2)^2 \left(-\frac{x}{2}\right)^3 + {}^5C_4 (3x^2)^1 \left(-\frac{x}{2}\right)^4 + {}^5C_5 \left(-\frac{x}{2}\right)^5$
2. ${}^nC_0 y^n + {}^nC_1 y^{n-1} \cdot x + {}^nC_2 y^{n-2} \cdot x^2 + \dots + {}^nC_n \cdot x^n$

Do yourself - 2

1. $\frac{70}{3} x^8$
2. $\frac{25!}{10! 15!} 2^{15} 3^{10}$
3. (a) -20 ; (b) $-560x^5, 280x^2$
4. (a) $\frac{5}{12}$; (b) $T_6 = 7$
6. ${}^nC_r (3^{n-r} - 2^{n-r})$
7. A, C, D
8. (a) $r = 6$; (b) $r = 5$ or 9
9. $\frac{17}{54}$
10. $n = 2$ or 3 or 4
11. 2

Do yourself - 3

1. 4th & 5th i.e. 489888
2. $n = 4, 5, 6$
3. (a) $T_7 = \frac{7 \cdot 3^{13}}{2}$; (b) 455×3^{12}
4. $\frac{5}{8} \leq x \leq \frac{20}{21}$

Do yourself - 4

1. C
3. $\frac{(2^{mn} - 1)}{(2^n - 1)(2^{mn})}$
8. (a) $\binom{30}{15}$
9. C
10. D

Do yourself - 5

1. -272160 or $-{}^{10}C_5 \times {}^5C_2 \times 108$
2. (a) 990 ; (b) 3660
3. (i) 3^n ; (ii) 1 ; (iii) a_n
4. (a) -1260 . $a^2 b^3 c^4$; (b) -12600

Do yourself - 6

2. 1
3. 001
5. 1
7. (i) 4 ; (ii) 3, 03, 803
8. 3

Do yourself - 7

1. $\frac{15015}{16}$
2. (i) 142 ; (ii) -197
3. (D)
4. (C,D)

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	B	A	A	C	B	A	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	D	D	C	B	A	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	A	B	B	B	B	D	A	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C,D	A,C,D	A,B	A,C	A,C,D	A,C	B,C,D	A,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C,D	C,D	A,C	A,C	C,D	A,C	A,C	A,B,C	A,B	B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	A	B	A,B	A,D	A,B	63.00	128.00	93.00	A
Que. 10	A	B	C	D					
	R	S	Q	P					

EXERCISE # O-4

1	2	3	4	5	6	7	8	9	10
0	2	2	0	6	12.00	500.00	-22100.00	-144.00	816.00

EXERCISE # JEE-MAIN

1. 3

2. Bonus

Note : In the problem 'number of terms should be 13 instead of 28', then (1) will be the answer.

- | | | | | | |
|--------|-------|-------|---------|--------|---------|
| 3. 1 | 4. 3 | 5. 4 | 6. 2 | 7. 3 | 8. 3 |
| 9. 1 | 10. 4 | 11. 1 | 12. 4 | 13. 4 | 14. 2 |
| 15. 1 | 16. 4 | 17. 2 | 18. 30 | 19. 3 | 20. 4 |
| 21. 51 | 22. 2 | 23. 1 | 24. (C) | 25. 57 | 26. 102 |

EXERCISE # JEE-ADVANCED

1	2	3	4	5	6	7	
C	8	5	646	6.20	A,B,D	3	